

POLS 641

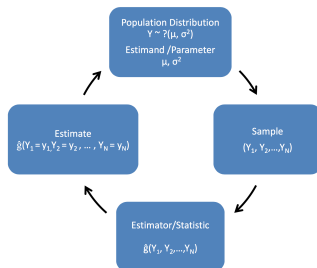
Introductory Political Analysis

Class 5: Measurement

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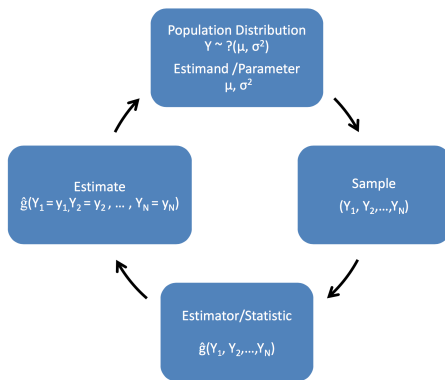
Warm-Up



An Important Research Question:

What is effect of coffee consumption on graduate student productivity?

Statistical Inference Process



How do we design research to use the best estimator?



Ingredients

150g unsalted butter, plus extra for greasing

150g plain chocolate, broken into pieces

150g plain flour

1/2 tsp baking powder

1/2 tsp bicarbonate of soda

200g light muscovado sugar

2 large eggs

Method

1. Heat the oven to 160C/140C fan/gas 3. Grease and base line a 1 litre heatproof glass pudding basin and a 450g loaf tin with baking parchment.

2. Put the butter and chocolate into a saucepan and melt over a low heat, stirring. When the chocolate has all melted remove from the heat.



estimand

estimator

estimate

Problem set # 1

Congratulations!



Housekeeping

Problem set #1:

- “Scores” posted on BB
- Grading
 - Check minus, Check, Check Plus
 - No personalized feedback(or come talk to me in office hours)
- I can see R output(check pdf as you go)
- careful with the term causal
- Fully respond to the questions(Get comfortable talking about stats!)
- Always correctly label figures
- Any questions?

Housekeeping

- Problem Set #2
 - No Rproject file. Use full file path(working directory) to load data
 - Practice knitting pdf early and if troubles come talk to me
 - Due next Wednesday
- Datacamp due tomorrow
- Final Project Questions? - Research Questions due next Friday
- Next week RIA - Support for White Nationalism

Today's Agenda

- Reviewing Random Variables
- Concepts and Measurement
- Univariate Analysis - Describing a single Variable
- Measures of Central Tendency

Random Variables (RVs)

We get a new piece of information (e.g. flip a coin, roll a dice, or measure someone's height). This is called:

- a random draw, or
- a random experiment

For any random experiment, the thing we are measuring is called a “random variable”.

Key properties of random variables:

- It has “a set” of possible values it can take (e.g. $\{heads, tails\}$, $\{1, \dots, 6\}$, or a positive number).
- Those outcomes have different probabilities of occurring. These are characterized by a distribution.
- You don't know which of those it will take until you do the random experiment/draw/observing.

Expectation and Variance

Again, random variables are measurements drawn from a distribution, whether we know that distribution or not.

Two important features of any distributions are **expectation** and **variance**.

- Expectation: the best guess about what number will be drawn from the distribution
- Variance: how far the numbers you draw tend to be from that best guess.

We will analyze the expectation and variance for a number of distributions, starting with normal.

Final words on normal distributions

In two weeks we'll see why the normal distribution appears so often.

For now, you should know that a normal distribution is completely described by *only* its expectation and variance.

Shift to thinking in terms of mean **AND** variation

Samples as Collections of Random Variables

Normally we have a *sample*: multiple RVs all with the same distribution. E.g.

- person 1 has $height_1$, a RV drawn from $N(\mu, \sigma^2)$
- person 2 has $height_2$, a separate RV drawn from $N(\mu, \sigma^2)$
- ...
- person N has $height_N$, yet another RV drawn from $N(\mu, \sigma^2)$

Even though each observation is a different RV, since they all come from the same distribution, we leverage the sample to learn something about that distribution.

The Sample Mean

The sample mean of $X_1, X_2, \dots, X_N$ is:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i = \frac{1}{N} [X_1 + X_2 + \dots + X_N]$$

Important notes:

- This is an example of an *estimator*: a function you can compute with data. You can use these estimators in R, for example `mean(object)`.
- **The mean is itself a random variable!** It takes on some particular value in your experiment, but it could have taken different values
- We are trying to understand what the *distribution of the mean* must look like, despite seeing the mean only once.

Why have we done all this?!

Remember, the big idea was that even when we only observe a mean once we can say a lot about how the mean would be *distributed* (as if we could observe it over and over again).

huge part of the statistical machinery depends on this finding. In two we'll see how this allows us to:

- Construct “confidence intervals” around your estimated mean to characterize our uncertainty
- hypothesis tests, get p-values, etc. for one-sample and two-samples tests involving means and eventually stars
- Other hypothesis tests for categorical outcomes, binary outcomes, counts, etc. will follow similar patterns.

Why have we done all this?!

- Takes on values in a range and with the probabilities defined by the distribution
- A random variable X is a (real-valued) function that maps the sample space to the real numbers (i.e. the function assigns a numerical value to each possible outcome of the experiment)
 - We define a random variable by its **Expectation** and **Variation**
 - $E[X]$ is almost always measured as the mean Variance is almost always measured as the standard deviation(more on this later)

On a final note, this is tough stuff but paves the way for much of what we will do, so go over these slides.

Questions?

Key Terms From Last Time

- 1 Population
- 2 Sample
- 3 Random Variable
- 4 Distribution(probability distribution)
- 5 Probability Density Function
- 6 Normal Distribution
- 7 Estimand
- 8 Estimator
- 9 Estimate
- 10 Expectation

Measurement

Operationalization and Describing Single Variables

Agenda

- Concept to Measurement
- Unit of Analysis
- Reliability and validity
- Descriptive Statistics
- Measures of Central Tendency
- Variation

Concepts

- Our description (and inference) is based on accurately measuring the concepts we are interested in
- Concept -> Operationalization -> Measure
- What are some examples of key (contested?) concepts in you've discussed in your other classes?

Key Elements of Concepts

- must be concrete - must be clearly defined(true of all types of research)
- must vary
- what are some “concepts” you are interested in ?
- identifying new concepts that are important and describing their variation is a great way to start a research project!

Key Elements of Concepts

- A conceptual definition clearly describes the concepts measurable properties and specifies the units of analysis to which the concept applies
 - specifies units of analysis
 - suggests the potential for variation along a constitutive set of characteristics
 - provides a theoretical way of determining the degree to which units exhibit those characteristics (i.e. the concept must be measurable in theory and hopefully in practice)
- The challenge for us is being able to quantify complex concepts
 - (We focus in this class on quantification, but issue is equally important in qualitative research)

Pollock's Conceptual Definition Template

The concept of [CONCEPT NAME] is defined as the extent to which [UNIT OF ANALYSIS] exhibit the characteristic of [CHARACTERISTICS THAT CONSTITUTE CONCEPT]

- e.g. "Liberalism is the extent to which individuals support increased government spending for social programs"
- Can you think of other examples?

Gerring Concept

Table 3.1: Criteria of Concept Formation

Resonance (familiarity, normal usage; *antonyms*: idiosyncrasy, neologism, stipulation)

- How faithful is the concept to extant definitions and established usage?

Internal coherence (depth, essence, fecundity, fruitfulness, natural kinds, power, real, thickness)

- How many attributes do referents of a concept share?

External differentiation (context, contrast-space, perspective, reference-point, semantic field)

- How differentiated is a concept from neighboring concepts? What is the contrast-space against which a concept defines itself?

Theoretical utility

- What utility does a concept have within a larger theory and research design?

Consistency (*antonym*: slippage)

- Is the meaning of a concept consistent throughout a work?

Key Term: Unit of Analysis

- Unit of Analysis: the entity we want to describe and analyze (individuals? groups? states? world orders?)
 - Surveys - Individuals, Voters, Households
 - Wars - States, state-dyad, state-years, dyad-years
 - Institutions - States, Local Governments, Representatives
- Unit of Observation: item that you actually observe while trying to
 - Often the same as Unit of Analysis, but not always
 - ie May aggregate individuals opinions(unit of observation) to say something about groups (unit of analysis)
 - ie want to say something about country behavior, but unit of observation is country-year

Key Term: Unit of Analysis

- can lead to miscommunication - ie Waltz's neorealist theory of foreign policy controversially claims not to predict the foreign policies of individual states
- **ecological inference fallacy** stems from when researchers try to make inferences about individuals from aggregate units of analysis
 - e.g. States with higher percentages of foreign born residents have higher levels of literacy

Operationalizing Concepts

- an operational definition describes the instrument to be used in measuring the concept and putting a conceptual definition "into operation" (ie we use an estimator to put the estimand "into operation")
- the conceptual definition should point the way toward constructing an operational definition
- e.g. liberalism is measured as a score of the number of times the respondent favored increasing government spending in a survey
- i.e. finding the appropriate estimator

Example: Operationalizing Concepts

- Hypothesis: Countries with IMF loans will experience more political instability than countries without IMF loans.
- What is meant by “political instability”?
- Defining the concept: Political instability means riots or mass demonstrations related to national politics.
- Operational definition: Any event with more than 1000 participants and a publicly stated goal related to national politics.

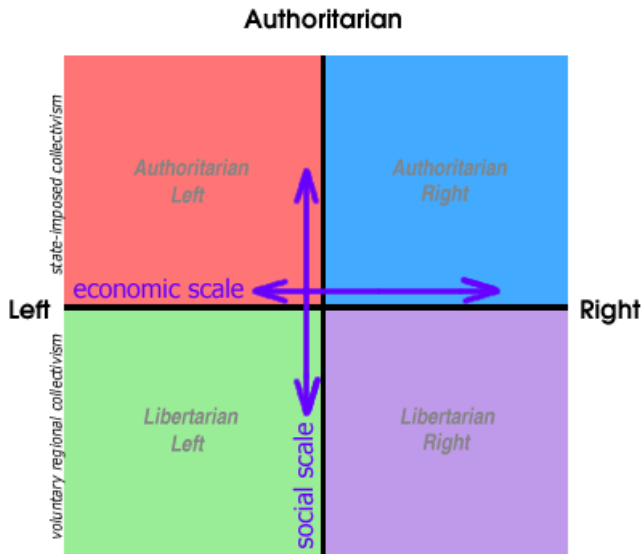
Operationalizing Concepts

- Hypothesis: Higher rates of literacy make democracy more likely.
- Defining concepts: Literacy: “the completion of enough education to be expected to read.”
Democracy: “a system of government in which public officials are selected in competitive elections.”
- Operational definitions: Literacy: “the % of people in a country that have completed at least 6 years of formal education.”
Democracy: “a country where the 2nd place finisher for the most powerful political office received at least 25% of the vote.”

Multidimensional Concepts

- a *conceptual dimension* is defined by a set of concrete traits of a similar type
- a *multidimensional concept* has more than one dimension
- e.g. "liberalism" as we usually understand it in American politics has both an economic dimension as well as a social dimension
- what are some other multidimensional concepts and what are their dimensions?
- Multidimensional concepts are often measured using index variables (ie measures that combine multiple indicators). However the weighting of each indicator is usually fairly arbitrary.

Multidimensional Concepts



Multidimensional Concepts - Grading Rubrics

Points	
Introduction (thesis statement and road map) and Conclusion	15
Application of Collective Action Theory	15
Application of alternative theory 1	15
Application of alternative theory 2	15
Argument Structure and Organization	10
Evidence/Use of Sources/Citations	10
Overall Writing Quality	15
Topic Selection	5
Total:	100

Measurement Error

- *systematic measurement* error introduces consistent, chronic distortion into an empirical measurement (i.e. bias)
- *random measurement error* introduces haphazard, chaotic distortion into the measurement process
- if you measure again, random error should be different, but systematic error will remain the same
- increasing sample size can help reduce random error, but NOT systematic error
- systematic error is a big problem, random error not so much
- Missing data - systematic error

Measurement Error

Certified Website of President
Donald J. Trump

Official 2020 Trump vs Democrat Poll

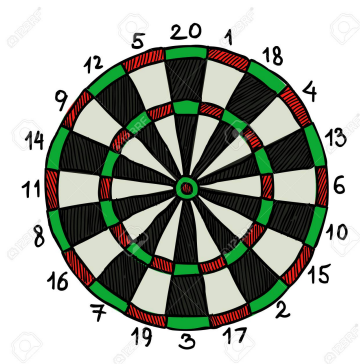
1. Who would you rather see fix our Nation's shattered immigration policies?
 - ☐ President Trump
 - ☐ A MS-13 Loving Democrat
2. Who do you trust more to protect America from foreign and domestic threats?
 - ☐ President Trump
 - ☐ A Corrupt Democrat
3. Who would you rather handle our Nation's economy?
 - ☐ President Trump
 - ☐ A Radical Socialist Democrat
4. Who do you believe is more transparent with the American People?
 - ☐ President Trump
 - ☐ A Lying Democrat



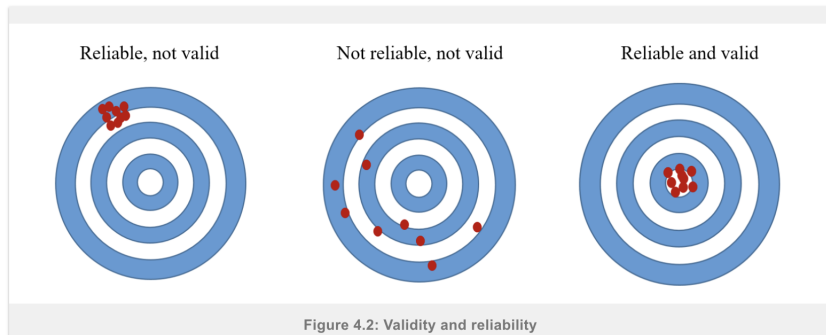
Reliability

- *reliability* refers to the degree to which a measure is consistent in measuring the concept
- To what extent would our measure yield the same results if we went out and collected more data?
- The more consistent the results, the higher the reliability
- Example: “Will you vote for Biden in 2024?” versus “on a scale from 0 (negative) to 100 (positive), what is your opinion of Biden?”
Second question is likely to be less reliable.

Validity and Reliability



Validity and Reliability



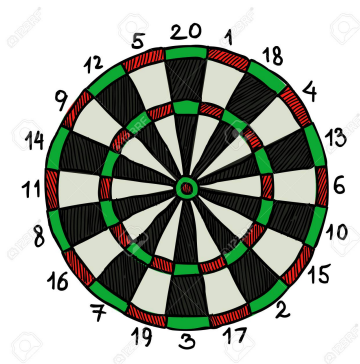
Validity

- *validity* refers to the degree to which the measure captures the "true" value of a characteristic
- Two Types
 - Internal
 - External(aka generalizability)
- If my measure repeatedly classifies Ted Cruz as a liberal, it may be reliable, but not valid.
- *Which form of validity would lab experiments be better at?*

Evaluating Validity

- Construct validity: Check to see if the measure is related to other measures thought to capture the same concept.
Example: Freedom House democracy scores should be related to POLITY democracy scores.
- Inter-item association: If we are combining measures thought to capture the same concept, all of these measures should be related.
Example: a set of survey questions on issues thought to capture liberalism should all be related.

Validity and Reliability



Validity and Reliability

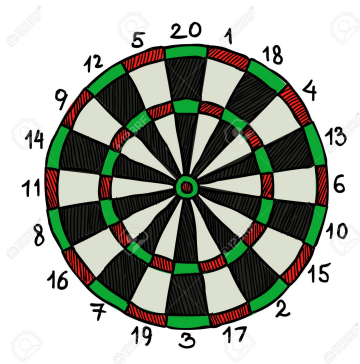
Evaluating Reliability

- test-retest method
- alternative-form method
- split-half method
- Cronbach's alpha

Evaluating Validity

- face validity
- construct validity

Validity and Reliability



A valid and reliable measure is a bullseye every time.

Terminology: Types of Variables

- 1 Nominal/Categorical: Discrete groups without an order
Factor or String data structure in R
- 2 Ordinal: Ordered but without consistent distance
- 3 **Continuous/interval:** Ordered and consistent distance
Numeric or integer data structure in R
- 4 Ratio

Terminology: Types of Data

- 1 Cross-sectional - information on units at a single point in time
- 2 Time-Series - a unit or few units observed at multiple points in time
- 3 Panel - information on the same units at multiple points in time(aka time-series, cross-sectional)

From Concept to Measurement

- ① Our statistical analysis is only as good as the data we put into it!!
 - ① You will spend more time thinking about and debating operationalization than modeling, theory, etc(in my experience)
- ② Valid and reliable operationalizations/estimators are essential to any research project(this is true of qualitative work too!)
- ③ Critiquing and improving upon existing measures is an opportunity to make a contribution
- ④ Remember “Not everything that can be counted counts, and not everything that counts can be counted” - William Cameron

Description vs Inference

- Our goal as social scientists is usually inference
 - i.e. A generalizable relationship between two variables
 - focus is on the relationship $X \rightarrow Y$
- Description
 - In unknown or novel populations, description itself can be finding.
 - State of the world i.e. polling or introducing a new dataset
 - Even if inference is the goal, need to be able to provide accurate information on what you are analyzing

Descriptive Statistics: Summarizing the data

Measures of Central Tendency

Describing the middle of the variable

- Mean
- Median
- Mode

Mean

- the total sum of all the numbers in a data set, divided by the total number of data points
- Most common form of central tendency
- Avoid the "Bill Gates walks in a bar . . ." problem

Median

- the middle of the data
- put in order and find the middle
- less influenced by extreme scores
- often, but not always, a better measure of central tendency than the mean.

Mode

- Most commonly occurring value
- Can be informative when discrete choices (ie survey questions)

Dispersion

Remember: a random variable is defined by a sample mean and standard deviation.

In other words, we need to think in variables in terms of both their central tendency and their distribution.

- 1 Variance
- 2 Standard Deviation(Standard Error)
- 3 Range
- 4 Interquartile Range/Percentiles

Variance

$$\text{Variance: } \sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

(We discussed this in Thursday's R workshop)

Standard Deviation: key idea

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

The standard deviation is the square root of the variance

Standard Deviation: key idea

Conceptually, the average dispersion, or the average distance from the mean.

Very similar to standard error, but different in important ways

Most common way we report dispersion

Other Ways to Look at Dispersion

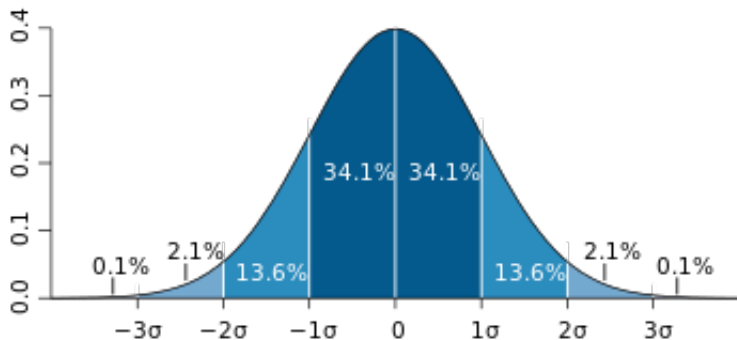
- Range
- Inter-quartile Range
- Outliers

Central Limit Theorem

- The mean from a sufficiently large number of independent draws from any distribution will be normally distributed
 - Symmetrical
 - Mean, Median, and Mode at some point
 - asymptotic tails
- the proportion of scores is consistent across standard deviations
- next week we'll talk about how this leads to ***

A Normally Distributed Variable

If we know $\bar{x}$ and σ we can calculate percentages of the sample in a given range



Z Scores and Z Distributions

A z score is a measure of how many standard deviations a particular observation is above or below the mean.

We subtract the mean from the observation and divide by the standard deviation.

z-Score Formula

$$z = \frac{x - \mu}{\sigma}$$

Questions?

Key Terms- Concept to Measurement

- 1 Concept
- 2 Operationalize
- 3 Unit of Analysis
- 4 Multidimensional Concept
- 5 Measurement Error
- 6 Reliability
- 7 Validity
- 8 Continuous
- 9 Categorical
- 10 Ordinal
- 11 Binary

Key Terms - Descriptive Stats

- 1 Central Tendency
- 2 Mean
- 3 median
- 4 Mode
- 5 Variance
- 6 Standard Deviation
- 7 Dispersion

Next Week

Hypothesis Testing

Relationships between variables

- Bailey - Ch 4
- Vox P-values (on Blackboard)
- How to get better at embracing unknowns
- Seeing Theory

Don't forget to complete the datacamp assignment